Title:

Anti-Light-Variation, Detailed Camera-Based Tactile Sensing With Continuous Binary Encoding

Authors:

Jinnuo Zhang[1], Jieji Ren[1], Zhirui Zhang[2], Yutong Pei[1], Zhenle Liu[1], Hao Shi[1], Ningbin Zhang[1], Boxin Shi[2], *Senior Member, IEEE,* and Guoying Gu[1], *Senior Member, IEEE.*

Affiliations:

[1] State Key Laboratory of Mechanical System and Vibration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; Shanghai Key Laboratory of Intelligent Robotics, Shanghai Jiao Tong University, Shanghai 200240, China

[2] State Key Laboratory of Multimedia Information Processing and National Engineering Research Center of Visual Technology, School of Computer Science, Peking University, Beijing 100871, China

Corresponding author: Guoying Gu (Email: guguoying@sjtu.edu.cn)

Contents

Supplementary Tables 3
Table S1. Analysis of changes in sensitivity metrics for FEM mesh size selection. 3
Table S2. Analysis of changes in sensitivity metrics for fringe width selection. 3
Table S3. Simulation parameter settings. 3
Table S4. Geometry reconstruction comparison for the fringe pattern under different sphere diameters (simulation model vs. real-data fine-tuned model) 4
Table S5. Geometry reconstruction results under diverse internal lighting interference 4
Table S6. Force estimation robustness under different lighting conditions. 5
Table S7. Feature analysis of automated needle insertion 5
Table S8. Hardware parameters of camera-based tactile sensors. 5
Supplementary Figures 6
Fig. S1. Sensitivity analysis and evaluation for fringe width selection. 6
Fig. S2. Mechanical parameter calibration based on deformation pattern matching 6
Fig. S3. Matching robustness comparison under significant deformation. 6
Fig. S4. Architecture of the proposed UNet with Attention for depth map estimation. 7
Fig. S5. Geometry reconstruction of complex everyday objects using the grid pattern. 7
Fig. S6. Geometry reconstruction under diverse ambient lighting interference. 8
Fig. S7. Internal illumination design of the sensor. 9
Fig. S8. Geometry reconstruction under diverse internal lighting interference 9
Fig. S9. Force estimation performance using the grid pattern. 10
Fig. S10. The design of ambient lighting interference for force estimation. 10
Fig. S11. Illumination robustness of force estimation. 11
Fig. S12. Comparison of geometry reconstruction between simulation training and real-data fine-tuning. 11
Fig. S13. Fluid property estimation. 12
Fig. S14. Robustness analysis of peak detection algorithm for automated needle insertion. 12
Fig. S15. Open-loop grasping experiment with an empty bottle. 13
Fig. S16. Relationship between geometry reconstruction accuracy and contact region. 13
Fig. S17. Durability test of FringeTac. 13
Fig. S18. Qualitative comparison of different camera-based tactile encoding schemes. 14
Fig. S19. Challenges of photometric reconstruction in camera-based tactile sensors. 14
Fig. S20. Comparison of shape reconstruction using marker-based encodings versus continuous binary encoding. 15
Fig. S21. Geometric feature resolution at a fixed pattern spacing (1.04mm). 15
Fig. S22. Object classification and spectrum analysis. 16
Supplementary Texts 17
Section S1. Parameter selection of fringe pattern 17
Section S2. Sensitivity analysis of mesh size selection 18
Section S3. Sensitivity analysis of fringe width selection 19
Section S4. Gel deformation simulation 20
Section S5. Mechanical parameter calibration 21
Section S6. Network for geometry reconstruction 22
Section S7. Illumination robustness of geometry reconstruction 22
Section S8. Illumination robustness of force estimation 22
Section S9. Object classification 23
Section S10. Robustness analysis of peak detection algorithm in automated needle insertion 23
Section S11. Fluid property estimation 24

Supplementary Tables

Table S1. Analysis of changes in sensitivity metrics for FEM mesh size selection.

Mesh size Δ (mm)	0.111	0.122	0.135	0.149	0.165	0.183	0.202	0.223	0.247	0.273
Convergence error ε (%)	1.67	-0.85	-2.98	-6.02	23.6	22.05	82.61	69.41	64.74	1544.63
Efficiency η	1.37	1.82	2.46	3.31	4.49	6.13	8.24	11.09	15.07	20.35

Note. Convergence error ε and relative computational efficiency η evaluated across 10 candidate finite element mesh sizes Δ. The reference baseline mesh size is set to $\Delta_0 = 0.100\text{mm}$. Detailed discussion on convergence bounds is provided in **Section S2**.

Table S2. Analysis of changes in sensitivity metrics for fringe width selection.

R (%)	W (mm)	$A(W)$	$I(W)$	$A'(W)$	$I'(W)$
5	2.11	0.9500	0.0475	0.0233	-0.0225
10	1.04	0.9000	0.0961	0.0945	-0.0923
20	0.51	0.8000	0.1945	0.3824	-0.3783

Note. Accuracy gain $A(W)$, tactile information gain $I(W)$, and their respective marginal derivatives $A'(W)$ and $I'(W)$ evaluated at candidate decay rates $R \in \{5\%, 10\%, 20\%\}$. The minimum resolvable fringe width is $W_{\min} = 0.100\text{mm}$. Detailed trade-off analysis is provided in **Section S3**.

Table S3. Simulation parameter settings.

Type	Parameter	Explanation	Value	Variation
Mechanical	E	Young's modulus	120 kPa	[80 kPa,160 kPa]
	μ	Friction coefficient	0.8	[0.6, 1.0]
	ν	Poisson's ratio	0.43	Fixed
	ρ	Density	1.0 g/cm^3	Fixed
Simulation	$h_{\max}$	Max pressing depth	2 mm	[1.5 mm, 2.5 mm]
	v_{press}	Pressing speed	25 mm/s	Fixed
	n_{step}	Simulation steps	9	Fixed

Table S4. Geometry reconstruction comparison for the fringe pattern under different sphere diameters (simulation model vs. real-data fine-tuned model).

Sphere Diameter (mm)	Mean $RMSE_{eff}$ (mm)	
	simulation model	real-data fine-tuned model
5	0.1198	0.0386
6	0.2501	0.0493
9	0.1647	0.0750
10	0.2280	0.1086

Table S5. Geometry reconstruction results under diverse internal lighting interference.

Internal Lighting condition		Mean $RMSE_{eff}$ (mm)
Color	Intensity	
Baseline		0.085
Red	Weak	0.122
Red	Strong	0.120
Green	Weak	0.085
Green	Strong	0.087
Blue	Weak	0.125
Blue	Strong	0.121
RGB	Weak	0.086
RGB	Strong	0.082
Random	Random	0.128

Note. Baseline is internal uniform white light.

Table S6. Force estimation robustness under different lighting conditions.

Lighting condition			RMSE (N)		
Ambient	Internal		F_x	F_y	F_z
	Color	Intensity			
Baseline			**0.04**	**0.07**	**0.27**
Fixed	Fixed		0.05	0.10	0.23
Dark	Fixed		0.05	0.10	0.28
Bright	Fixed		0.04	0.08	0.39
Dynamic	Fixed		0.05	0.08	0.30
Fixed	Random	Random	0.05	0.14	0.61

Note. The baseline uses indoor natural ambient light and uniform white internal illumination.

Table S7. Feature analysis of automated needle insertion.

Experiments	Time (s)	Peak value (px)	Speed (px/s)
1	3.6	5.3	1.47
2	2.6	6	2.31
3	3.1	5.8	1.87
4	2.7	4.6	1.70
All	3.0±0.5	5.4±0.6	1.84±0.35

Table S8. Hardware parameters of camera-based tactile sensors.

Sensors	Pixel Size (mm/px)	Sensing Area (mm^2)	Image Resolution (px×px)
TacTip	0.0640	~1257	640×480
GelSlim2.0	0.0625	1200	640×480
GelSlim3.0	0.0469	675	640×480
GelSight	0.0286	252	640×480
DelTact	0.0380	675	798×586
Ours	**0.0417**	**535**	**640×480**

Note: Pixel size represents the physical length corresponding to one image pixel, calculated as $\sqrt{A/(W \times H)}$, where A is the sensing area (mm^2), and $W \times H$ is the image resolution (px × px). The unit is mm/px.

Supplementary Figures

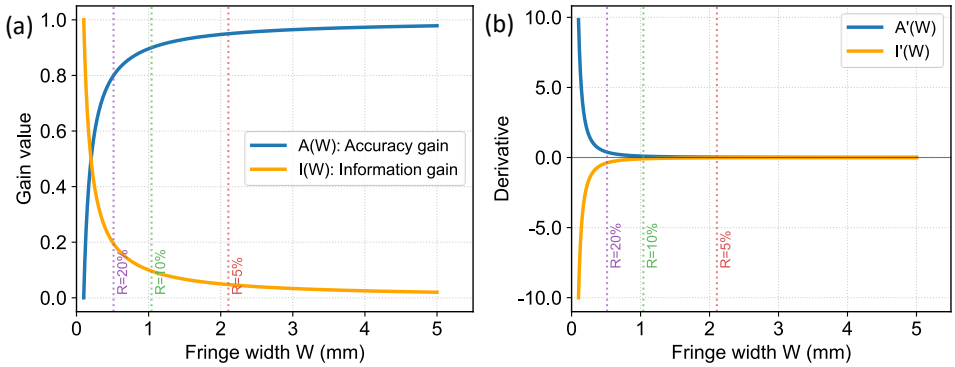


Fig. S1. Sensitivity analysis and evaluation for fringe width selection.
(a) Normalized accuracy gain $A(W)$ and tactile information gain $I(W)$ plotted against fringe width W. Vertical dotted lines denote candidate points corresponding to decay rates R = 20% (W = 0.51mm), R = 10% (W = 1.04mm), and R = 5% (W = 2.11mm). (b) Marginal derivatives $A'(W)$ and $I'(W)$. The marginal benefit of accuracy has essentially approached zero when the decay rate R < 10%.

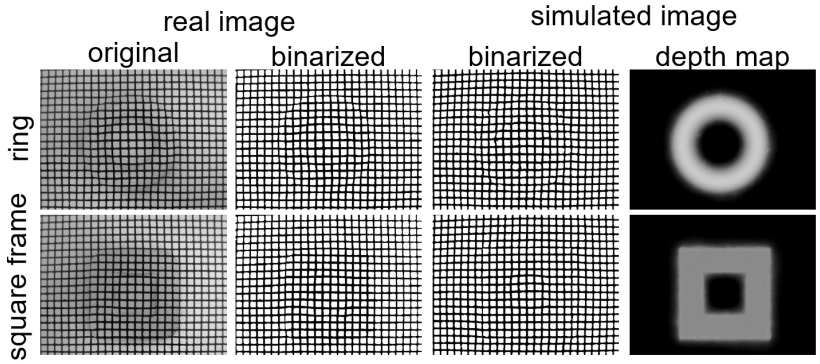


Fig. S2. Mechanical parameter calibration based on deformation pattern matching.
Comparison of physical and simulated indentation results for two rigid indenters using the optimized gel parameters. The high consistency between the real and simulated binary patterns under optimal parameters validates the fidelity of the calibrated simulation baseline.

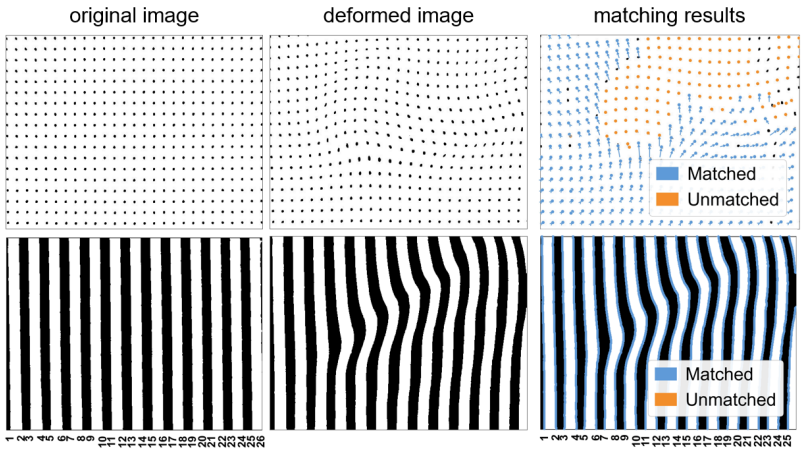


Fig. S3. Matching robustness comparison under significant deformation.
The top row shows optical flow tracking results on conventional discrete markers. Under large contact deformation, discrete markers suffer numerous severe mismatching and losses. The bottom row shows tracking results on the proposed continuous binary fringe pattern with the same spacing as the markers under equivalent large-deformation conditions. Since stripe patterns provide continuous features that are easier to recognize than marker points, 100% accuracy is maintained when tracking boundary lines. All 26 fringe boundaries are successfully matched, shown in blue solid curves, demonstrating superior physical robustness under extreme deformation.

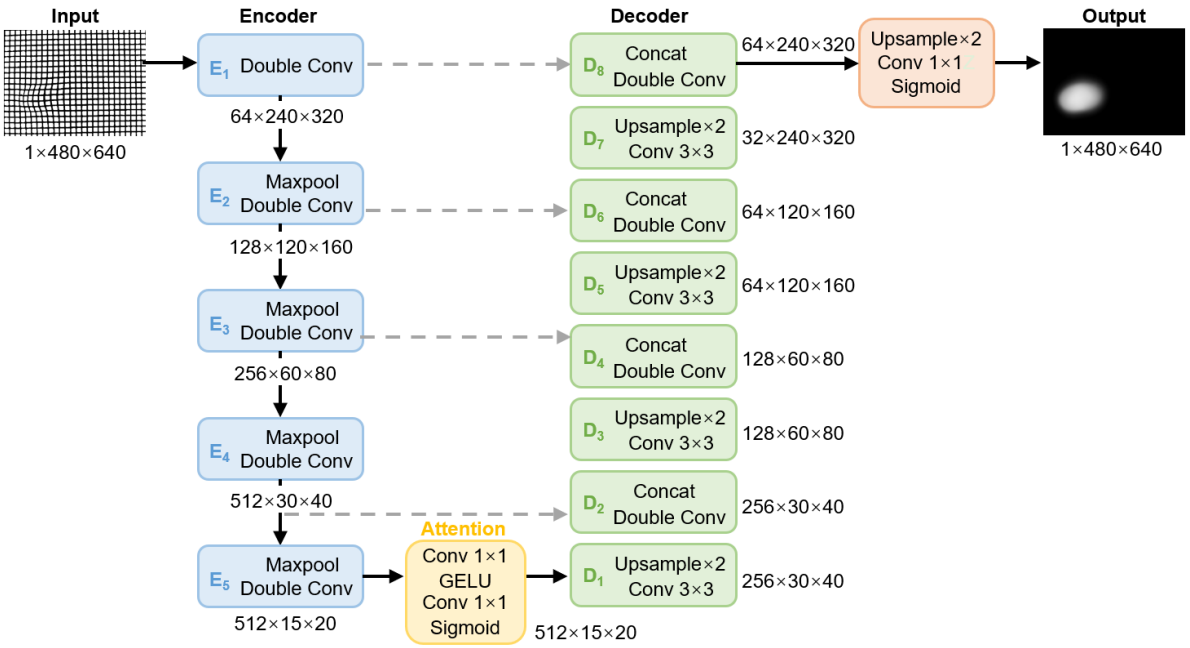


Fig. S4. Architecture of the proposed UNet with Attention for depth map estimation.
The network takes a binary image ($1\times480\times640$) as input and predicts the corresponding depth map ($1\times480\times640$). For the encoder subnetwork, the initial block E_1 downsamples the input to $64\times240\times320$, matching the feature dimensions required by the subsequent residual-like downsampling blocks E_2-E_5. For the channel attention module positioned at the deepest bottleneck layer, features are first compressed via a 1×1 convolution with GELU activation, and subsequently re-expanded using the Sigmoid activation to compute channel-wise attention weights. For the decoder subnetwork and output layer, feature maps are progressively upsampled and concatenated with their corresponding encoder features via skip connections. The final representation from D_8 ($64\times240\times320$) is bilinearly upsampled in the output layer before being mapped through a 1×1 convolution and a Sigmoid activation to output the target depth map.

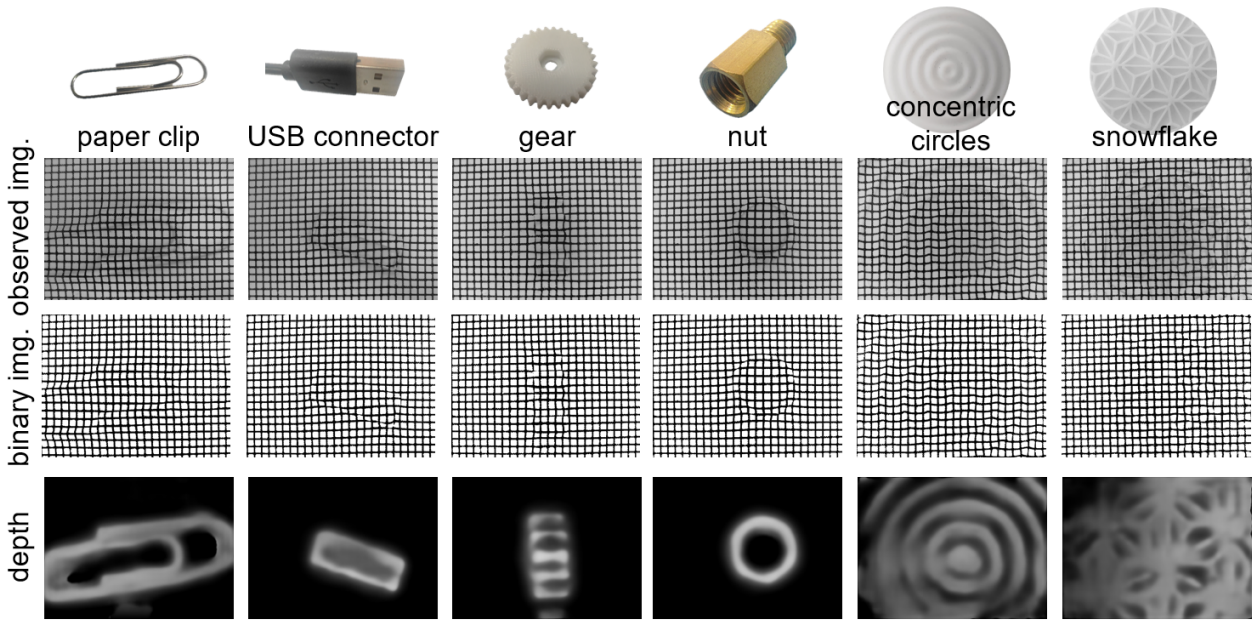


Fig. S5. Geometry reconstruction of complex everyday objects using the grid pattern.

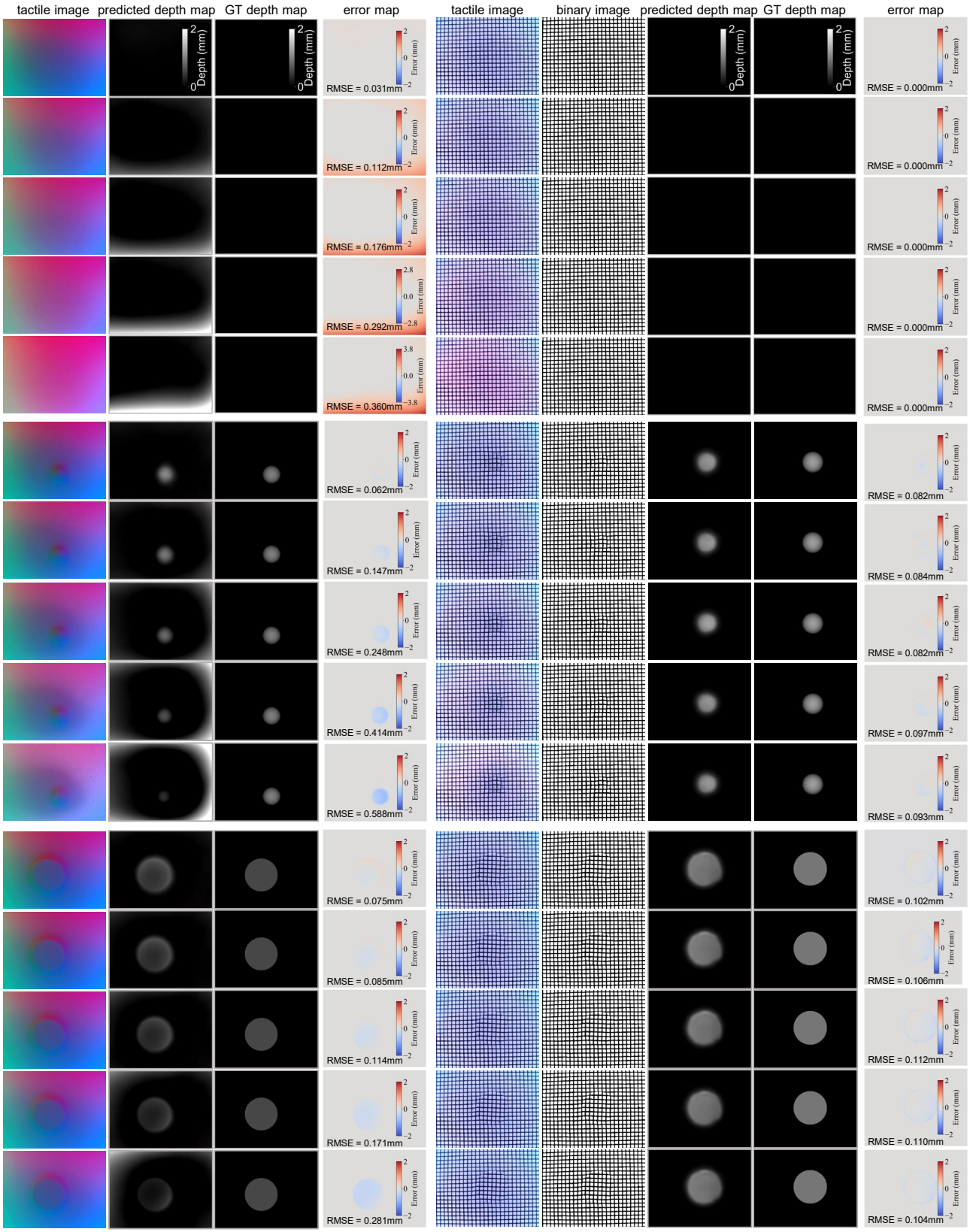


Fig. S6. Geometry reconstruction under diverse ambient lighting interference.
Geometry reconstruction comparison between the PS method (left) and the grid-pattern-based continuous binary encoding method (right) under different ambient illumination disturbances (red, white, and blue lights across interference levels 0-4). Each block displays partial results for three contact states from top to bottom: unpressed state (red light interference), spherical indenter (white light interference), and cylindrical indenter (blue light interference).

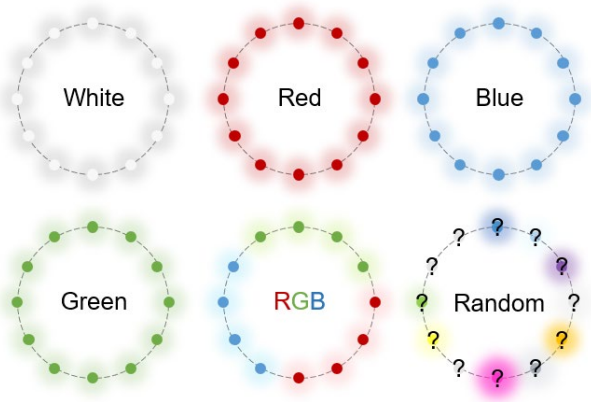


Fig. S7. Internal illumination design of the sensor.
6 illumination conditions are designed for the internal lighting. In addition to the initial white-light illumination, five interference conditions are designed. Specifically, RGB refers to evenly dividing the internal LEDs into three groups to provide red, green, and blue illumination, respectively. Random illumination means that the color and intensity of each LED vary randomly. For the first four interference conditions, two illumination intensities, weak and strong, are further considered.

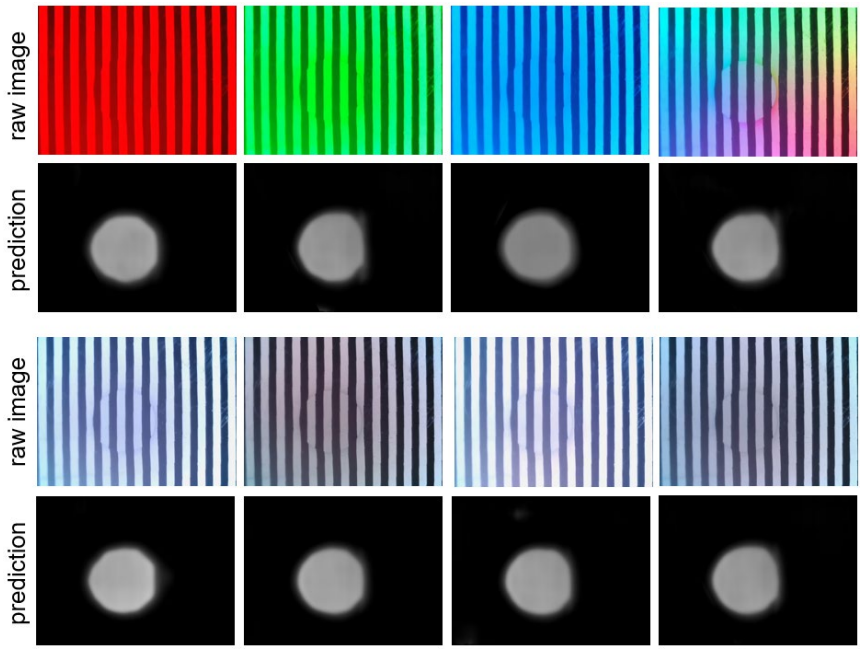


Fig. S8. Geometry reconstruction under diverse internal lighting interference.
Under a constant indentation state, the upper two rows show the observed images and corresponding reconstructed depth maps under strong red, green, blue, and RGB internal illumination. The lower two rows show results under random internal illumination (random intensities for each RGB channel of each LED). Geometry reconstruction remains robust across all tested environments.

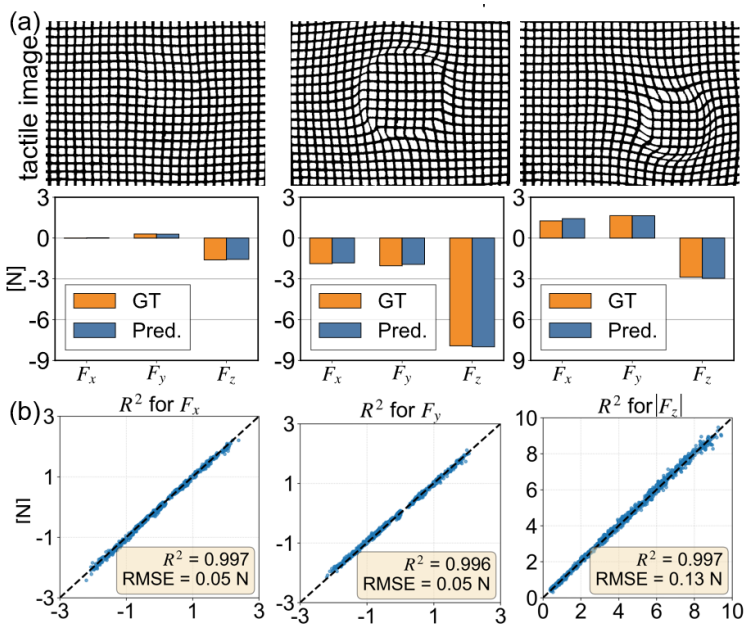


Fig. S9. Force estimation performance using the grid pattern.
(a) Representative tactile images with corresponding ground truth and predicted force vectors F_x, F_y, F_z. (b) Regression analysis (R^2 plots and RMSE metrics). Using the same data collection procedure and network architecture as for the fringe pattern, the 2D grid structure demonstrates great spatial isotropy with balanced tangential accuracy (RMSE = 0.05 N) for both F_x and F_y, while significantly improving normal force estimation accuracy (F_z RMSE reduced from 0.27 N to 0.13 N).

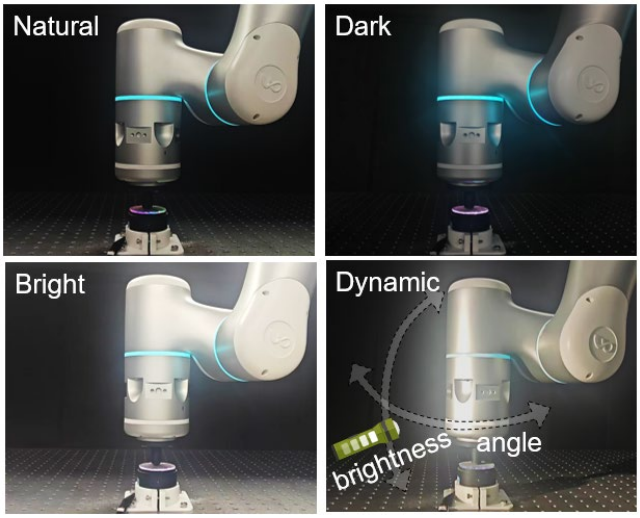


Fig. S10. The design of ambient lighting interference for force estimation.
Ambient lighting variations include four representative conditions: indoor natural light, darkness (lights off at night), bright illumination with additional light sources, and dynamically varying illumination. Dynamic ambient illumination is generated using a handheld flashlight whose intensity is switched among four discrete levels while its illumination direction is varied continuously.

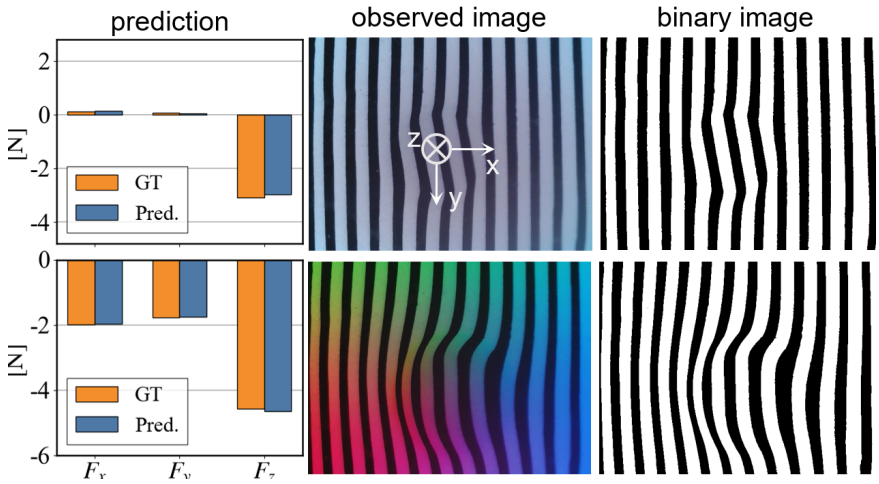


Fig. S11. Illumination robustness of force estimation.
According to the leftmost image, it can be seen that the network's predictions of F_x, F_y, and F_z closely match the corresponding ground-truth values when the lighting conditions randomly change, achieving robust force estimation.

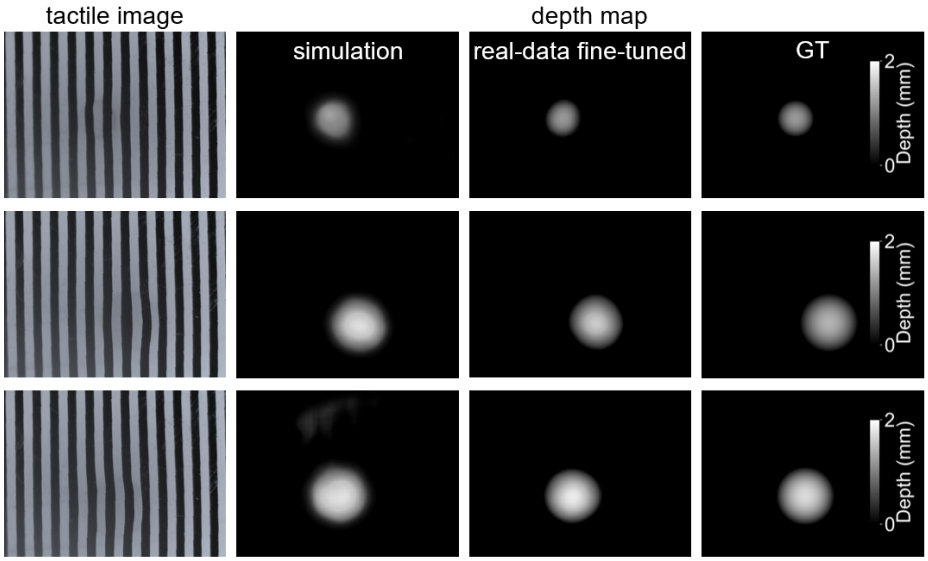


Fig. S12. Comparison of geometry reconstruction between simulation training and real-data fine-tuning.
Taking the fringe pattern as an example, each row displays from left to right: the input tactile image during contact with spherical objects of varying diameters, the depth map predicted by the network trained purely on simulation data, the depth map predicted by the network fine-tuned with a small dataset of real-world samples, and the corresponding ground truth depth map. Real data fine-tuning can effectively improve the reconstruction performance and make the predicted depth map closer to the ground truth value. This proves its effectiveness in narrowing the sim-to-real gap.

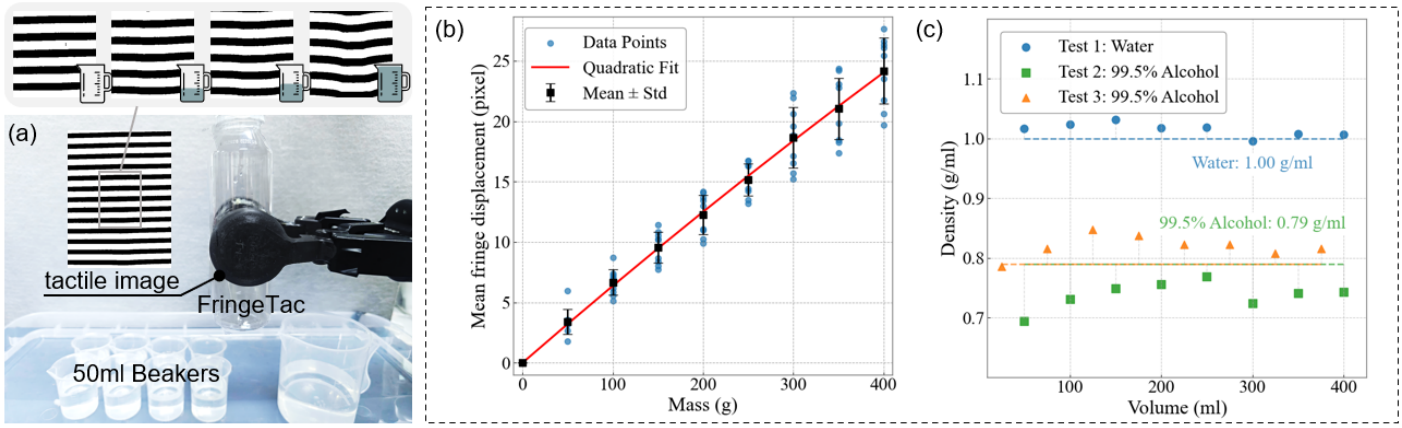


Fig. S13. Fluid property estimation.
(a) Water is incrementally added to the bottle to establish the relationship between mean fringe displacement and liquid mass. (b) The fitted model exhibits a near-linear correlation. (c) Based on this model, static density estimation is performed, while dynamic mass monitoring during continuous pouring is shown in Video S2.

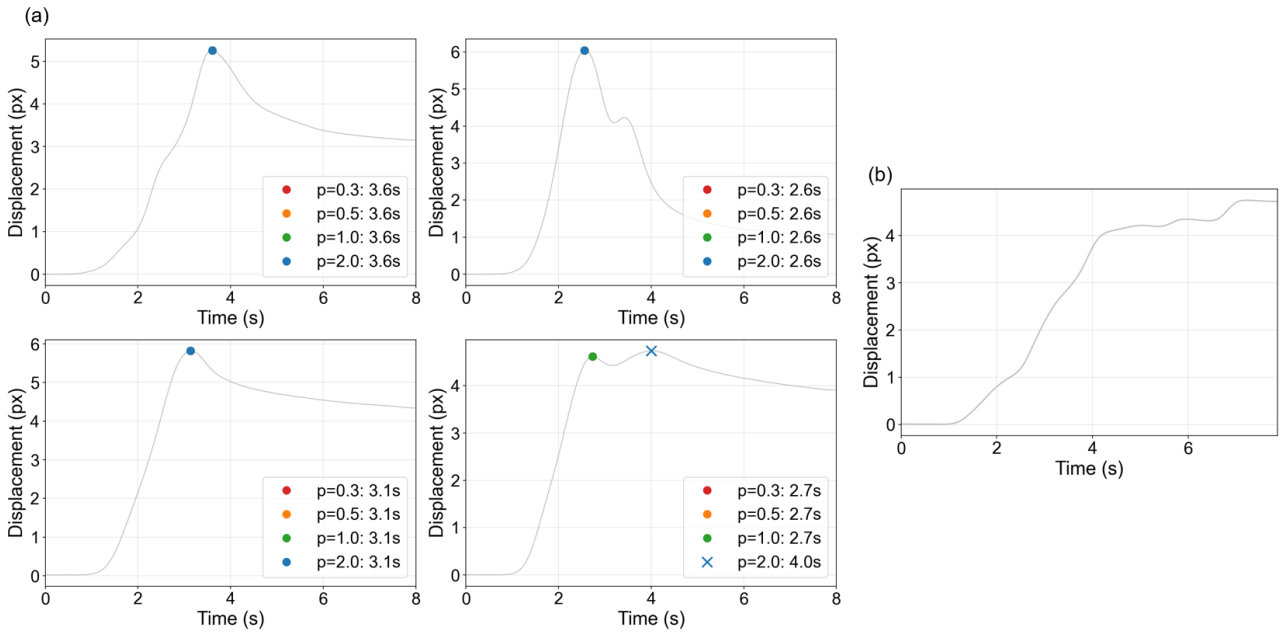


Fig. S14. Robustness analysis of peak detection algorithm for automated needle insertion.
(a) To evaluate the robustness of the initially selected threshold, we varied it by scaling factors of p = 0.3, 0.5, and 2.0, and re-detected the puncture peaks accordingly. Identical peak times detected under different thresholds confirm algorithm stability. A misidentification occurs only in Trial 4 at an excessively high threshold (p = 2.0). (b) An unsuccessful insertion trial. The syringe slipped while being held, preventing successful insertion into the vessel. Consequently, no puncture peak was detected because penetration did not occur.



Fig. S15. Open-loop grasping experiment with an empty bottle.
Without tactile feedback, the gripper is unable to modulate its closure force through pose adjustment. Nevertheless, owing to the low weight of the empty bottle, the initial grasping force is adequate for successful grasping in open-loop trials.

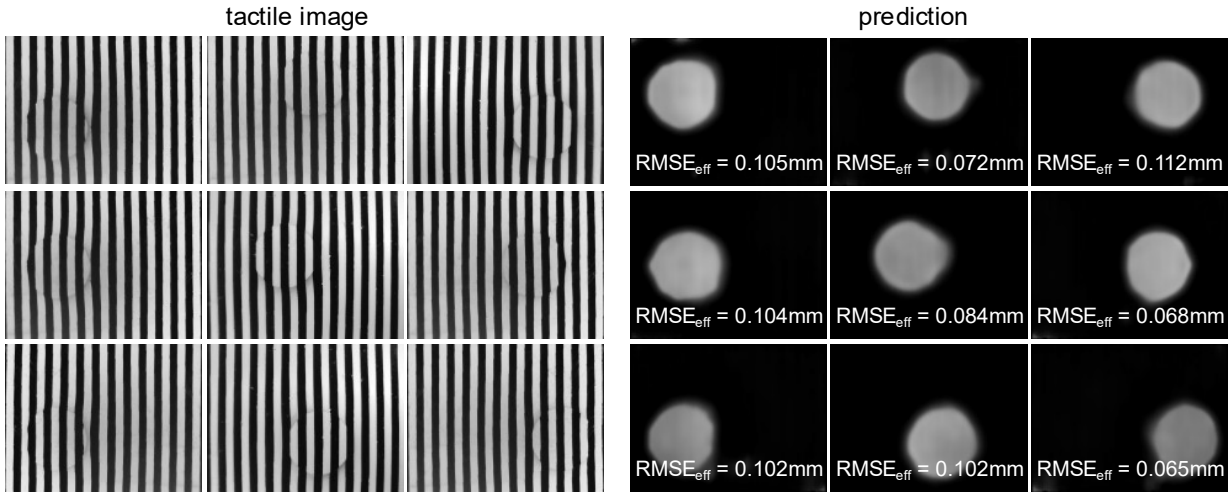


Fig. S16. Relationship between geometry reconstruction accuracy and contact region.
A cylindrical indenter is pressed to the same depth at various positions across the sensing area. Tactile images and the corresponding reconstructed depth maps are obtained, and the RMSE is computed against the ground truth. The results exhibit minor fluctuations in RMSE across different contact locations, all of which remain within a controllable range.

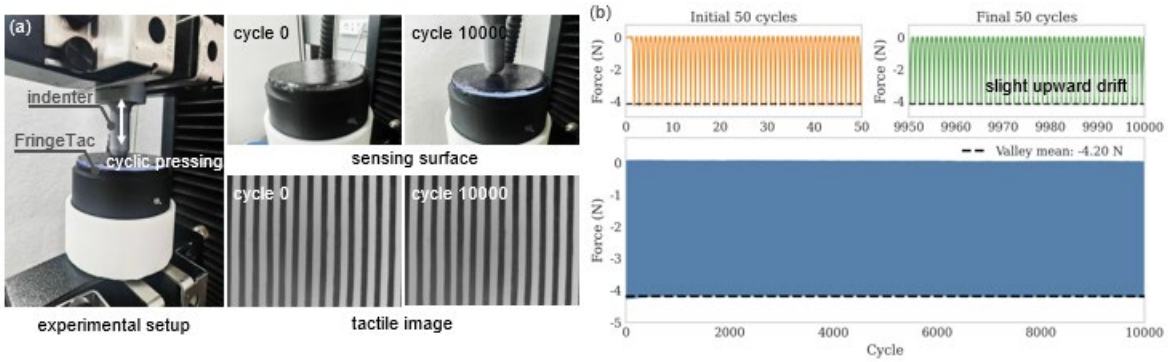


Fig. S17. Durability test of FringeTac.
A spherical indenter is pressed against the sensor surface to a fixed depth, performing a series of loading-unloading cycles. Over 10,000 consecutive loading–unloading cycles, the force response remains highly consistent, with only a slight upward drift. No visible plastic deformation or delamination of the elastomer gel is observed, and the binary fringe pattern shows no signs of degradation or fading. This robust structural resilience implies satisfactory long-term reliability for repeated, high-frequency robotic manipulation tasks.

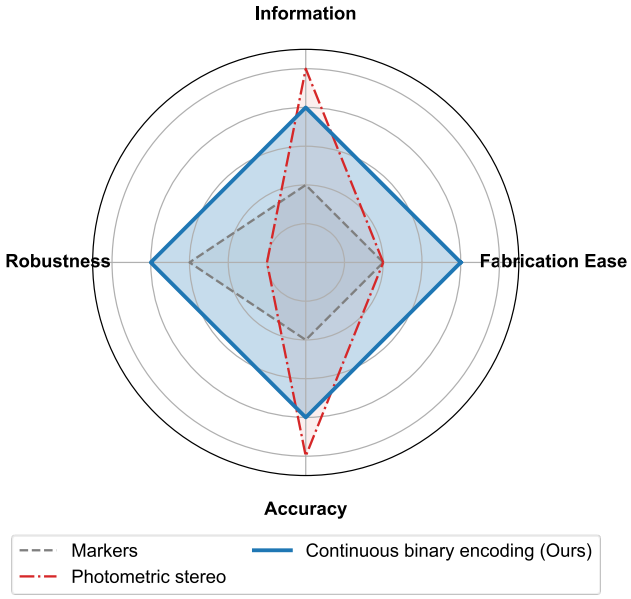


Fig. S18. Qualitative comparison of different camera-based tactile encoding schemes.
Performance is evaluated across four core dimensions: Fabrication Ease, Information, Robustness, and Accuracy. Robustness refers to both tracking robustness and low dependence on a carefully designed light field. Compared with traditional discrete markers and photometric stereo, the proposed continuous binary encoding achieves a well-balanced tradeoff, combining low dependence on complex illumination design with accurate reconstruction and sufficient tactile information.

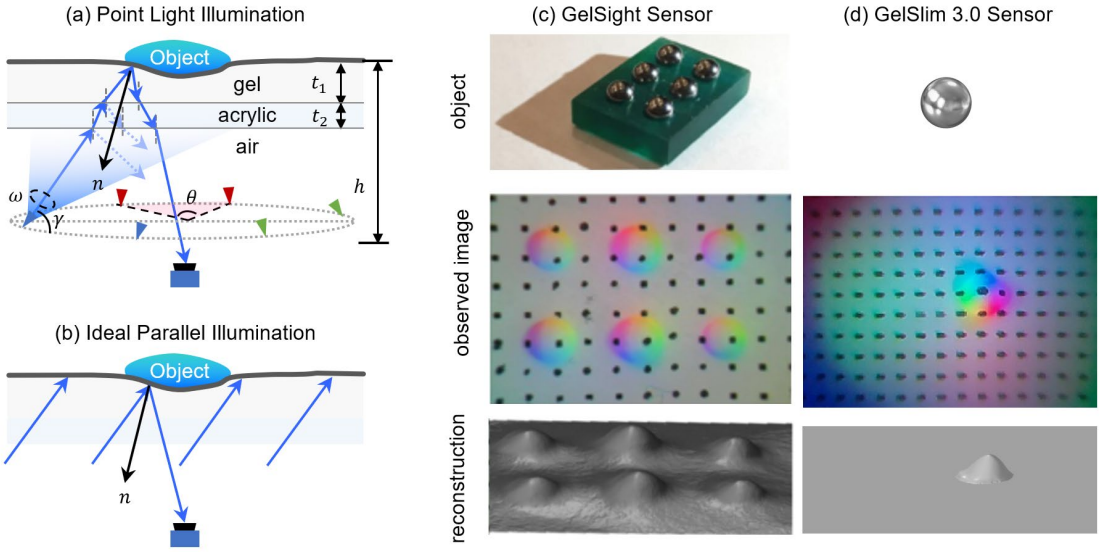


Fig. S19. Challenges of photometric reconstruction in camera-based tactile sensors.
(b) The idealized photometric stereo model assumes parallel illumination, where light rays are perfectly parallel, no shadow occurs and both intensity and incident angles remain constant across the sensing surface. (a) However, in real sensors, this ideal condition often cannot be met. The short distances between point light sources and the sensing surface cause near-field illumination, producing substantial variations in incident light direction and intensity. Refraction and reflection at the gel–acrylic–air interfaces further alter the light field, while contact deformation introduces shadows and self-occlusion. These effects increase the difficulty of illumination design and calibration, particularly in compact or nonplanar sensor configurations. (c, d) Representative spherical-contact reconstructions from GelSight and GelSlim 3.0. Both sensors have optimized illumination, with GelSlim 3.0 additionally employing a light-shaping lens. However, issues with non-uniform illumination persist. The displayed sphere reconstructions exhibit peaked surface profiles, showing the difficulty of light-field optimization.

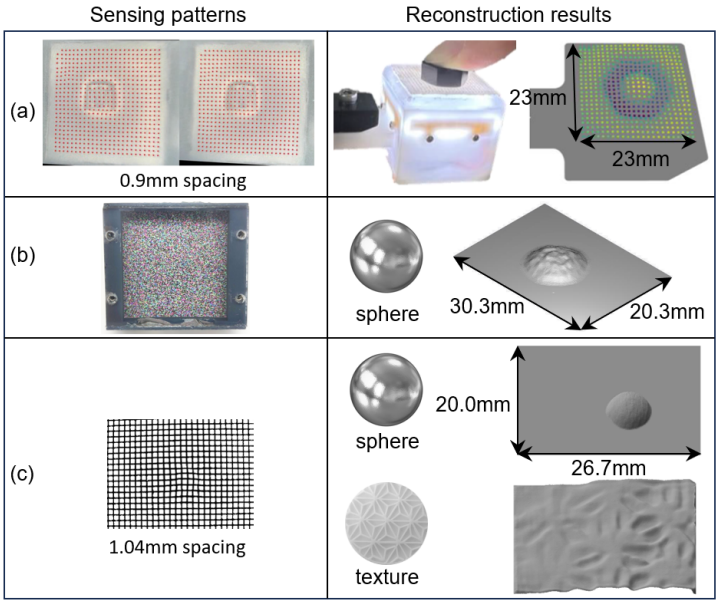


Fig. S20. Comparison of shape reconstruction using marker-based encodings versus continuous binary encoding.

(a) GelStereo 2.0 with a marker spacing of 0.9 mm. (b) DelTact with a dense color pattern. (c) Our continuous binary encoding with a pattern spacing of 1.04 mm. GelStereo 2.0 uses binocular vision to recover marker coordinates in three dimensions and reconstructs a sparse point cloud geometry. The sampling density is limited. DelTact instead obtains richer spatial measurements through dense color-pattern tracking. Although color optimization improves feature distinguishability, dense sampling places tighter demands on pattern fabrication and feature matching. Local nonuniformities are also visible in the displayed reconstructions. Our method uses continuous black-and-white boundaries to provide spatially connected deformation observations and constraints for stable feature extraction, producing detailed reconstructions with smooth surface transitions, as illustrated by the sphere and complex textures.

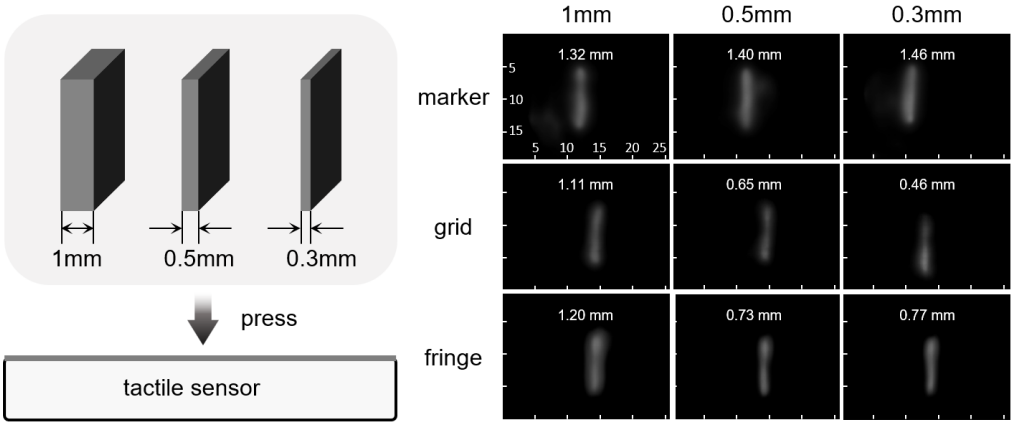


Fig. S21. Geometric feature resolution at a fixed pattern spacing (1.04mm).

Steel sheets with thicknesses of 1.0, 0.5, and 0.3 mm are repeatedly pressed edge-on into the sensing surface (N = 10) for geometry reconstruction under identical experimental conditions. Some examples of reconstructed depth maps and the corresponding thickness estimates are shown. The sheets are pressed along the short axis of the sensing field, parallel to the fringe boundaries in the fringe-pattern configuration. Considering weak deformation at the periphery of the contact area caused by gel compliance, small depth values are filtered out, and the central indentation region is selected. Local widths are measured perpendicular to the longitudinal axis of the indentation and averaged to estimate the sheet thickness. Marker-based reconstruction provides poor discrimination among the three thicknesses. Continuous binary encoding captures deformation variations along connected boundaries, providing richer spatial cues for resolving sub-millimeter features. The fringe pattern distinguishes the 1.0 mm sheet from the two thinner ones, and the grid pattern captures decreasing widths, successfully resolving all three thicknesses.

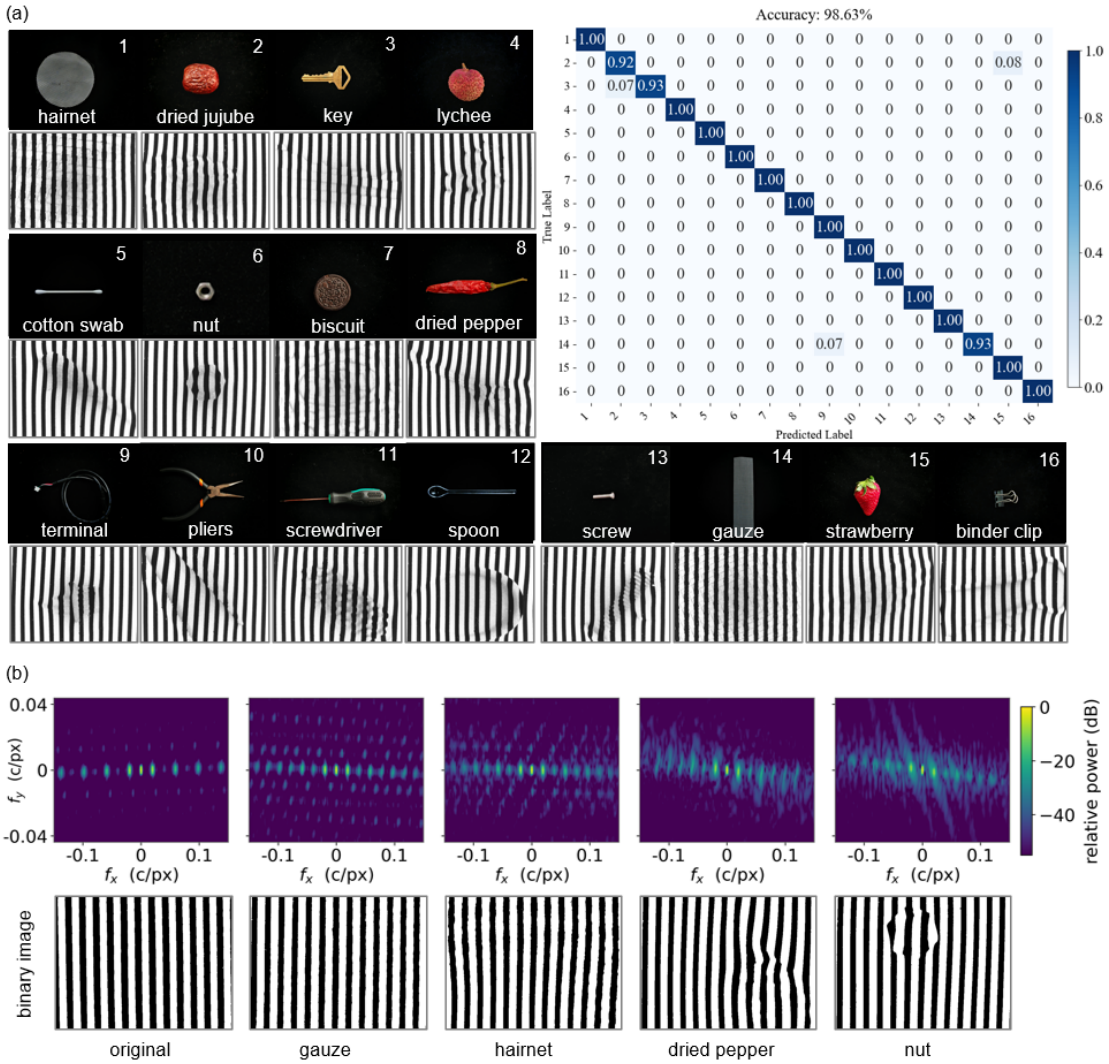


Fig. S22. Object classification and spectrum analysis.
(a) For each object, approximately 100 tactile images are collected. A model based on ResNet-18 is employed and the dataset is divided into training and test sets in a 9:1 ratio. The confusion matrix shows an overall accuracy of 98.63%, indicating good performance in recognizing diverse features. (b) To analyze the spatial features, we apply a two-dimensional fast Fourier transform (FFT) to the binary images of some representative examples to obtain their spatial spectra, which clearly show the obvious different features of diverse objects in the low-frequency (center) and high-frequency (peripheral) domains. The axes f_x and f_y denote horizontal and vertical spatial frequencies in cycles/pixel. Relative power (in dB) is normalized to the maximum power across the five cases, which is set as the 0-dB reference. For the original pattern, the fringe harmonics lie mainly along the horizontal midline (f_y =0), because the vertical fringes vary across x but remain nearly unchanged along y. Detailed textures such as gauze and hairnets produce more distinct structure at higher frequencies, whereas coarser contact features such as dried peppers and nuts produce more diffuse distributions at lower frequencies with weaker harmonic peak contrast.

Supplementary Texts

Section S1. Parameter selection of fringe pattern

Since the geometry reconstruction network is trained on simulated data, the fringe parameters must be compatible with the spatial resolution of the finite element mesh. In practice, both the simulation mesh size and the fringe width jointly determine how faithfully fringe deformation can be represented. As a result, parameter selection is carried out in two stages. We first identify a mesh resolution. The fringe width is then selected under this fixed resolution based on the shape representation error we defined.

As expected, dense meshes yield higher geometric accuracy, but at a substantial computational cost. Coarser meshes, on the other hand, significantly reduce runtime while introducing noticeable discretization errors. To quantify this trade-off, mesh sizes Δ ranging from 0.1mm to 5mm are evaluated, uniformly sampled in logarithmic space, as shown in **Fig. 3**(e) (only $[0.1\text{mm}, 3\text{mm}]$ is shown in the figure).

Depth maps generated using the finest mesh are treated as reference, and the mean absolute error (MAE) of reconstructions obtained with coarser meshes is computed accordingly. For each mesh size, three independent simulations are performed and averaged to suppress stochastic variation.

We take the natural error (MAE) fluctuation between the two densest grid resolutions as the reference accuracy level. Mesh sizes whose MAE stays within $\pm 10\%$ of this reference are regarded as acceptable, as they provide comparable reconstruction quality with substantially reduced computational burden. Following this criterion, the largest acceptable mesh size is selected as $\Delta^* = 0.149\text{mm}$.

With the mesh size fixed at Δ^*, the fringe width W becomes the dominant factor controlling encoding fidelity. Finer fringes provide denser tactile information, but when the fringe scale approaches the mesh scale, the triangular elements can no longer accurately approximate the deformation of fringe boundaries, resulting in amplified shape distortion. **Fig. 3**(d) shows the shape error approximating fringe boundaries with discretized meshes. To isolate the effect of fringe width, we fixed the duty cycle $D = 0.5$.

For a fringe of width W, let A_i denote the area of the i-th triangular element that intersects the fringe region. The total mesh area required to fully cover the fringe is then given by $A_{\text{sum}} = \sum_i A_i$, where the summation includes all triangular elements whose support overlaps with the fringe. Due to mesh discretization, this accumulated area is generally larger than the ideal fringe area $W \times 1$.

Based on this observation, we define the relative fringe shape representation error as

$$E_{\text{shape}} = \frac{A_{\text{sum}} - W}{W}, \tag{S1}$$

which directly measures the relative area discrepancy between the mesh-based approximation and the ideal fringe geometry. Intuitively, this quantity reflects how efficiently the mesh can represent a fringe of a given width. For consistency, all evaluations are conducted on fringe segments of fixed unit length, so that the ideal fringe area is $W \times 1$. This normalization eliminates the influence of fringe length and allows the analysis to focus exclusively on how fringe width affects geometric representability.

We evaluate W in the range from 0.1mm to 5mm, again sampled uniformly in logarithmic space, as shown in **Fig. 3**(f) (only $[0.1\text{mm}, 3\text{mm}]$ is shown in the figure). It is worth noting that E_{shape} is inherently a function of W. To mitigate position-dependent discretization bias, three fringe segments placed at different spatial locations are evaluated for each fringe width, and the resulting E_{shape} are averaged. Repeating this procedure for all tested widths yields a relationship $E(W)$ between W and E_{shape}

$$E(W) = 0.167W^{-0.983} + 0.152. \tag{S2}$$

We further examine how rapidly the shape error decays as the fringe width increases. A normalized relative decay rate is introduced,

$$R(W) = \frac{E(W) - E_{\infty}}{E(W_{\min}) - E_{\infty}} = \left(\frac{W}{W_{\min}} \right)^{-0.983}. \quad \text{(S3)}$$

This function satisfies $R(W) \in (0,1]$, decreases monotonically with W and quantifies the proportion of remaining reducible error relative to the smallest fringe width. As with the acceptable range of 10% taken for mesh size optimization, the fringe width corresponding to $R(W) = 10\%$ is also selected, resulting in $W = 1.04\text{mm}$.

Finally, given the minimum resolvable fringe width, equal widths for black and white fringes maximize effective tactile information under a fixed spatial frequency. The duty cycle is therefore set to $D = 0.5$.

Section S2. Sensitivity analysis of mesh size selection

We conducted a mesh convergence study to determine the optimal finite element mesh size. We define the computational efficiency metric η and the relative convergence error ε as follows:

$$\eta = \left(\frac{\Delta}{\Delta_0} \right)^3, \quad \varepsilon = \frac{E - E_0}{E_0} \times 100\% \quad \text{(S4)}$$

Δ is the target mesh size, $\Delta_0 = 0.100\text{mm}$ represents the reference fine-mesh baseline. Since the computational time in 3D finite element analysis scales roughly with the cube of the mesh density, we define computational efficiency η as the factor by which the computation time at the current mesh size Δ is reduced compared to that of the finest mesh Δ_0.

E is the MAE evaluated at the current mesh size Δ, and E_0 is the error computed on the finest mesh Δ_0. The relative convergence error ε is defined as the percentage difference from the finest-mesh error. A candidate mesh size is considered converged once ε consistently settles within a specified threshold bound.

To thoroughly evaluate the sensitivity of convergence to different candidate threshold bounds (e.g., ε = 5%, 10%, 20%), we list the corresponding metrics for the 10 adjacent mesh sizes in the range of 0.111mm to 0.273mm, where convergence is approached. The corresponding convergence errors ε and computational efficiencies η are summarized in **Table S1**.

As shown in **Table S1**, when the mesh size exceeds 0.165mm, the convergence error sharply spikes above 20%, indicating clear non-convergence due to under-sampling of localized deformations. Conversely, when the mesh size is reduced below 0.149mm, the error rapidly drops and remains bounded within a narrow range.

Evaluating the choice of convergence threshold yields the following insights:
Under a stricter 5% threshold, the candidate mesh size is $\Delta = 0.135\text{mm}$ (efficiency $\eta = 2.46$).
Under a slightly relaxed threshold of 10%, the candidate mesh size increases to $\Delta = 0.149\text{mm}$ (efficiency $\eta = 3.31$).

Overall, both 0.135mm and 0.149mm represent viable choices depending on the chosen threshold. In our implementation, 0.149mm is adopted based on two practical considerations. First, its convergence error ($\varepsilon = -6.02\%$) lies very close to the 5% threshold boundary while providing a noticeable 34.5% enhancement in computational efficiency ($\eta = 3.31$ vs. 2.46). Second, as the mesh size decreases past 0.149mm, the error transitions from large positive values into a regime of minor positive and negative oscillations around zero. Therefore, we infer that the mesh enters the stable asymptotic convergence range starting at 0.149 mm.

Section S3. Sensitivity analysis of fringe width selection

Selecting the optimal fringe width W involves a fundamental trade-off between accuracy and tactile information density. On one hand, increasing the fringe width improves spatial sampling per fringe, reducing shape representation error. On the other hand, a smaller fringe width allows a higher number of fringes within the camera's field of view (FOV), thereby maximizing the total effective tactile information captured.

In **Section S1**, we derived a fitting function $E(W)$ relating the shape representation error to the fringe width:

$$E(W) = 0.167W^{-0.983} + 0.152 \tag{S5}$$

We also defined a normalized relative decay rate $R(W)$ to describe the proportion of residual error relative to that at the minimum resolvable fringe width $W_{\min} = 0.100\text{mm}$:

$$R(W) = \frac{E(W) - E_{\infty}}{E(W_{\min}) - E_{\infty}} = \left(\frac{W}{W_{\min}} \right)^{-0.983} \tag{S6}$$

To select an appropriate R threshold for width selection, we conduct a sensitivity analysis. We define two normalized, monotonic gain metrics bounded in $[0,1]$:

$$A(W) = 1 - R(W) \tag{S7}$$

$$I(W) = \frac{W_{\min}}{W} \tag{S8}$$

where accuracy gain $A(W)$ quantifies the achieved reconstruction accuracy improvement and information gain $I(W)$ represents the amount of effective information contained in the pattern within the fixed FOV. $I(W)$ is directly proportional to fringe density or in other words, inversely proportional to fringe width W.

To evaluate the sensitivity of threshold selection, we analyzed the performance metrics $A(W)$, $I(W)$, and their corresponding marginal derivatives $A'(W)$ and $I'(W)$ across candidate decay rates $R \in \{5\%, 10\%, 20\%\}$, as detailed in **Table S2** and **Fig. S1**.

As illustrated in **Fig. S1**, increasing W continuously decreases the information gain ($I'(W) < 0$ universally) and increases the accuracy gain $A(W)$. However, the accuracy gain $A(W)$ exhibits diminishing marginal returns as W increases.

At $R = 20\%$ ($W = 0.51\text{mm}$), the information gain $I(W) = 0.1945$, and the derivative of the precision gain is $A'(W) = 0.3824$, indicating that significant marginal gains still exist.

At $R = 10\%$ ($W = 1.04\text{mm}$), the derivative $A'(W)$ drops to 0.095. The marginal benefit of further increasing the fringe width is no longer significant.

At $R = 5\%$ ($W = 2.11\text{mm}$), accuracy gain reaches 95%, but at a severe cost to spatial pattern density ($I(W)$ drops to 0.0475). Furthermore, the derivative $A'(W)$ drops to 0.0233, confirming that further widening the fringes yields negligible accuracy improvements.

Therefore, $W = 1.04\text{mm}$ ($R = 10\%$) is finally selected as the optimal fringe width. Furthermore, comparing this fringe width to the optimal FEM element size ($\Delta = 0.149\text{mm}$), the ratio $W / \Delta \approx 7$ is close to an order of magnitude. This spatial resolution is basically sufficient to characterize complex fringe deformations.

Section S4. Gel deformation simulation

To simulate realistic gel deformation during contact, we adopt the TacIPC simulator as the framework backbone. This simulator integrates a finite element method (FEM) with Incremental Potential Contact (IPC), allowing stable and accurate modeling of hyperelastic gel materials under complex contact conditions.

As shown in **Fig. 4**(b), the gel is discretized into a tetrahedral mesh. The gel deformation at each time step is computed by minimizing the total energy of the system.

Specifically, the updated vertex positions $x^{t+\Delta t}$ are obtained by minimizing the physically motivated energy:

$$\begin{aligned} x^{t+\Delta t} &= \arg\min_x \ \mathcal{E}_{\text{mech}} + \mathcal{E}_{\text{cont}} + \mathcal{E}_{\text{fric}} \\ \mathcal{E}_{\text{mech}}(x, x^t, \mathbf{v}^t) &= \frac{1}{2} \| x - x^{\text{pred}} \|_{\mathbf{M}}^2 + \Delta t^2 \mathcal{W}(x) \\ \mathcal{E}_{\text{cont}}(x) &= \sum_{i \in \mathcal{P}} \eta \varphi(d_i(x)) \\ \mathcal{E}_{\text{fric}}(x, x^t) &= \sum_{i \in \mathcal{P}} \mu \cdot \lambda_i \cdot \psi(\| \mathbf{u}_i \|). \end{aligned} \tag{S9}$$

Here, the mechanical energy $\mathcal{E}_{\text{mech}}$ encodes the inertial and elastic dynamics, the contact barrier energy $\mathcal{E}_{\text{cont}}$ ensures non-penetration between gel and objects, and the frictional energy $\mathcal{E}_{\text{fric}}$ models the tangential friction resistance during surface contact.

Specifically, in $\mathcal{E}_{\text{mech}}$, $\mathbf{M}$ is the mass matrix, Δt is the time step size, and $\mathcal{W}(x)$ represents the internal strain energy derived from a hyperelastic material model, such as Neo-Hookean, parameterized by Young's modulus E and Poisson's ratio ν. The predicted position x^{pred} accounts for external force f_{ext} and the velocity v^t: $x^{\text{pred}} = x^t + \Delta t v^t + \Delta t^2 \mathbf{M}^{-1} f_{\text{ext}}$.

$\mathcal{E}_{\text{cont}}$ discourages the proximity of node-triangle or edge-edge beyond a specified distance threshold, where $\mathcal{P}$ is the set of active proximity pairs, $d_i(x)$ is the signed distance between surfaces in pair i, η is the penalty scaling factor, and $\varphi(\cdot)$ is a smooth logarithmic barrier function that diverges as $d_i \to 0$.

$\mathcal{E}_{\text{fric}}$ penalizes relative tangential sliding, where μ is the friction coefficient, $\mathbf{u}_i$ denotes relative tangential motion at contact i, $\psi(\cdot)$ is a smooth regularization function (e.g. pseudo-Huber) and λ_i is an approximation of the magnitude of normal force, often calculated as $\lambda_i = \eta \left\| \nabla \varphi(d_i) \right\|$.

To align simulated gel behavior with the real sensor, we first determine the mechanical parameters of silicone gel in the real world. Subsequently, we further calibrate and refine mechanical parameters by comparing simulated deformed patterns with real tactile images. Ring-shaped and hollow square objects are pressed against the real gel surface to induce distinctive deformation features. The same contact configurations are replicated in simulation under the same maximum indentation depth $h_{\max}$, pressing speed v_{press}, and simulation steps n_{steps}.

By varying Young's modulus E, friction coefficient f, we generate multiple simulated deformation states. Undeformed binary patterns are then mapped onto the simulated meshes to obtain deformed patterns, which are compared against real images using image similarity metrics. The final calibrated parameters are $E = 120\text{kPa}, \quad f = 0.8, \quad \nu = 0.43, \quad \rho = 1.0\text{g/cm}^3$.

Since ink wear and material aging mainly affect surface friction and stiffness, E and f are allowed to vary within a specified range during data generation, while ν and ρ are kept fixed. The simulation parameters are summarized in **Table S3**. For each pressing trial, mesh states from steps 5 to 9 are exported, representing different indentation depths. In total, 1205 pressing simulations are conducted over 33 objects, resulting in 6025 deformed gel meshes.

Section S5. Mechanical parameter calibration

To ensure high simulation fidelity, the material and contact interaction parameters of the elastomeric gel substrate are calibrated through physical indentation experiments. The density and Poisson's ratio of the silicone gel are fixed at empirically chosen values: $\rho = 1.0\text{g/cm}^3$ and $\nu = 0.43$. The Young's modulus E and the friction coefficient f between the gel surface and indenters are identified via grid-search optimization.

Physical calibration is performed using a sensor coated with a 2D grid pattern, as it provides high spatial tactile information across both orthogonal axes. Two custom 3D-printed rigid indenters (ring, square frame) are selected as the calibration objects. Both feature a closed-loop structure with an internal cavity and each combines planar features with curved boundaries, thereby providing a richer set of morphological characteristics.
Both indenters are modeled in CAD and placed under identical indentation conditions (identical contact locations and indentation depths) in both physical experiments and finite element simulations.

A total of $9 \times 15 = 135$ parameter combinations are evaluated across a two-dimensional search space:

E : $[0.04\text{MPa}, 0.20\text{MPa}]$ with a step size of 0.02MPa (9 candidates).

f : $[0.1,1.5]$ with a step size of 0.1 (15 candidates).

Evaluating these parameter sets across 2 calibration indenters yielded a total of 270 simulated deformation cases.

For each simulation case, the surface mesh displacement field is mapped to synthesize the corresponding deformed grid image. Both real physical images and simulated images are preprocessed via thresholding to extract binary grid patterns.

The image matching error is evaluated using the Mean Absolute Error (MAE) of normalized pixel intensities ($I \in [0,1]$):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left| I_{\text{sim}}(i) - I_{\text{real}}(i) \right| \tag{S10}$$

where N is the total pixel count within the sensor ROI. The image similarity score S is defined as the optimization fitting objective:

$$S = 1 - \text{MAE} \tag{S11}$$

where $S = 1$ indicates identical binary pattern alignment and $S = 0$ indicates complete mismatch.

By maximizing the joint average similarity score across both indenters, the optimal material and contact parameters are identified as:

$$E^* = 0.12\text{MPa}, \quad f^* = 0.8, \quad \nu = 0.43, \quad \rho = 1.0\text{g/cm}^3$$

Under these optimal parameters, the residual mismatch and similarity performance are:

Square frame: Similarity score $S = 0.7781$ ($\text{MAE} = 0.2219$).

Ring: Similarity score $S = 0.6743$ ($\text{MAE} = 0.3257$).

There are many possible reasons for the residual mismatch, like manufacturing tolerances (layer lines) of the 3D-printed indenter and minor numerical instabilities arising from a very small number of distorted mesh elements during computation. Therefore, after completing the initial calibration of mechanical parameters, we fine-tuned the network by incorporating a small amount of real data into the synthetic simulation dataset; this step aimed to reduce the gap between simulation and reality and optimize the reconstruction results.

Section S6. Network for geometry reconstruction

To achieve geometry reconstruction, we design a network that maps binarized images of FringeTac to depth maps. The network adopts a U-Net architecture. The network configuration is shown in **Fig. S4**. The input is a binarized image of size $640 \times 480 \times 1$, and the output is a single-channel depth map of identical resolution.

The training objective is to minimize a hybrid loss function that jointly considers global geometric accuracy and local edge preservation. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{edge}} \tag{S12}$$

Here, $\lambda = 0.05$ is a weighting coefficient, and $H = 480$, $W = 640$ are the height and width of the image, respectively. The first term $\mathcal{L}_{\text{MSE}}$ represents the pixel-wise mean squared error between the predicted depth map $\hat{Y}$ and ground truth.

$$\mathcal{L}_{\text{MSE}}(\hat{Y}, Y) = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} (\hat{Y}_{ij} - Y_{ij})^2 \tag{S13}$$

The second term $\mathcal{L}_{\text{edge}}$ is an edge-aware loss activated after 100 epochs, aiming to enhance depth discontinuities. It is computed as the absolute difference between the gradient magnitudes of the predicted and ground truth depth maps.

$$\mathcal{L}_{\text{edge}}(\hat{Y}, Y) = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \left| G(\hat{Y})_{ij} - G(Y)_{ij} \right| \tag{S14}$$

Section S7. Illumination robustness of geometry reconstruction

Partial reconstruction results are provided in **Fig. S6** to further illustrate how ambient light affects the spatial distribution of errors across the sensor surface.

For the photometric stereo (PS) method, monochromatic light (red, green, or blue) mainly interferes with the corresponding R, G and B channel of captured images. Because the internal LEDs illuminate the sensor surface from three directions spaced 120 degrees apart, the interference effect is most pronounced in the diagonal area farthest from its illumination source, resulting in a gradual decrease in reconstruction error from the periphery towards the center.

In contrast, the grid-pattern-based continuous binary encoding method processes binarized deformation features rather than raw multi-channel intensities. Consequently, its spatial error distribution remains uniform and largely invariant to external lighting changes across all contact states and interference levels.

To systematically evaluate illumination robustness, in addition to ambient light interference, we design different internal illumination interference conditions (**Fig. S7**). Because PS is inherently modeled on constant internal RGB intensity relationships, the model no longer holds when internal illumination changes. Therefore, we only test binary patterns, using the fringe pattern as an example (**Fig. S8**). We press a cylindrical indenter into the fixed sensor to a constant depth while capturing images under various illumination combinations (**Table S5**). We calculate the RMSE between the predicted depth map and the ground truth depth map. Across all tested illumination categories, the RMSE fluctuation range is small, indicating that the binarized patterns successfully preserve critical geometric information.

Overall, the continuous binary encoding scheme exhibits excellent overall illumination robustness.

Section S8. Illumination robustness of force estimation

Similarly to the case of geometry reconstruction, we evaluate the robustness of force estimation on the test dataset for different ambient (**Fig. S10**) and internal (**Fig. S7**) lighting interference conditions. The results are shown in **Table S6**. The standard dataset is collected under fixed ambient and internal illumination (indoor natural light and uniform white LEDs) for network training, while additional datasets under varied illumination are used to assess robustness without retraining. The RMSE of F_x exhibits high stability in all conditions tested. When only the ambient illumination varies while the internal lighting remains fixed, the estimation errors for all force components change slightly. When the internal illumination changes dynamically and randomly, the increase in errors is relatively large but remains within a controllable margin.

Section S9. Object classification

To evaluate the tactile perception capability of FringeTac, we perform an object perception task using 16 commonly encountered items. These objects represent three general categories: food, household items, and tools. Each object is assigned a numerical label from 1 to 16, as shown in **Fig. S22**. To assess the discriminative capacity of the model, we select several objects with similar surface textures, such as a hairnet and gauze (both mesh-like), strawberry, lychee, and dried jujube (all featuring irregular bumpy surfaces).

For each object, approximately 100 tactile images are collected by pressing the item onto the FringeTac surface. Among them, 90% are used for training and 10% for testing. A ResNet-18 model is employed as the classifier, with its input and output dimensions adapted for this task. The model is trained in an end-to-end manner and achieved a classification accuracy of 98.63% on the test set, indicating robust and accurate performance in recognizing diverse contact features.

Three types of misclassification are observed: dried jujubes are predicted as strawberries, keys as dried jujubes, and gauze as terminal connectors. The confusion between the dried jujube and the strawberry occurs because they share similar irregular surface profiles, leading to comparable deformed patterns.

The latter two errors are more likely attributable to variations in contact pose and partial occlusion during image acquisition. Since the tactile image only captures the region within the camera's field of view, if a distinctive feature (such as the serrated edge of a key) is not fully captured, the model may rely on secondary features that resemble those of another object (such as the uneven surface of a dried jujube). Similarly, when only a small region of the gauze is pressed, its sparse mesh pattern may resemble the localized texture of a terminal connector.

Section S10. Robustness analysis of peak detection algorithm in automated needle insertion

During the experiment, the sensor calculates the average fringe displacement $\Delta\bar{p}$ from the captured images and applies real-time filtering to suppress noise. Since $\Delta\bar{p}$ qualitatively reflects the magnitude of the resistive force, a notable drop in $\Delta\bar{p}$ occurs when the needle tip penetrates the first vessel wall. The system identifies this event by monitoring the displacement drop $\Delta\bar{p}_{\text{drop}}$ following a local maximum. Once $\Delta\bar{p}_{\text{drop}}$ exceeds a preset threshold, the penetration is detected and the insertion is halted. The characteristics of the data curve variations for all trials are summarized in **Table S7**.

The threshold is determined empirically from open-loop experiments. It must be set sufficiently high to avoid false triggers due to minor signal fluctuations, yet low enough to prevent missed detections. To evaluate its robustness, we applied scaling factors $p = 0.3$, 0.5, 2.0 to the original threshold and re-evaluated all trials, as shown in **Fig. S14(a)**. The detected penetration peaks remained consistent across nearly all trials. The only exception occurred in trial 4, where the excessively high threshold $p = 2.0$ caused the algorithm to misinterpret a post-peak fluctuation as a penetration event, confirming the overall robustness of our method.

Fig. S14(b) presents the displacement curve from the unsuccessful penetration trial. In this case, the suboptimal initial grasping posture of the syringe caused the needle tip to slip upon contact, preventing the axial force from reaching the required level for penetration. Accordingly, no peak is detected, and the trial is classified as a failed puncture.

Section S11. Fluid property estimation

Based on FringeTac, we conduct an experiment focusing on an open-loop tactile calibration task for fluid-related load perception, aiming to characterize the relationship between fringe displacement and grasped mass.

Using the high sensitivity of the sensor to tangential deformation, our goal is to infer the weight and indirectly the density of unknown liquids through tactile observation based on contact, without prior knowledge of the geometry of the container. As shown in **Fig. S13(a)**, FringeTac is integrated into one end of a robotic gripper mounted on a 6-DoF robotic arm. An empty plastic bottle is grasped with the fringe extension direction orthogonal to the bottle's central axis. Water is incrementally added in 50 ml steps, up to 400 ml. After each addition, a tactile image capturing fringe deformation is recorded.

We use the previously defined mean fringe displacement $\Delta\overline{p}$ (in pixels) as an indicator to measure changes in liquid weight. This averaged metric is then used to fit a regression model that captures the relationship between fringe displacement and liquid mass.

To minimize the influence of the initial gripping posture and the inherent weight of the empty bottle, the image captured immediately after gripping the empty bottle is designated as the initial image, and changes in $\Delta\overline{p}$ are calculated relative to this initial state. The experiment is repeated ten times. Nine trials are used to fit a quadratic model relating the liquid mass m (in grams) to fringe displacement:

$$\Delta\overline{p} = -0.000012m^2 + 0.0649m, \quad R^2 = 0.9992. \tag{S15}$$

As illustrated in **Fig. S13(b)**, fringe displacement increases in a near-linear manner with the grasped mass, with only mild nonlinearity at higher loads. This result confirms that fringe displacement provides a stable and sensitive proxy for tangential load variation during grasping.

Based on the fitted model, we further conducted static density estimation for different liquids and dynamic mass monitoring during continuous pouring. As illustrated in **Fig. S13(c),** FringeTac achieved low density estimation error for both water and 99.5% anhydrous ethanol. The average density estimation error of water (10th trial) is 0.0182g/ml, and the mean density estimation errors for 99.5% ethanol are −0.0517 g/mL and 0.0341 g/mL in Tests 2 and 3, respectively.

After completing the static density estimation experiment, we conduct a dynamic monitoring experiment. 400 ml of water is continuously and evenly poured into the bottle, and the changes in liquid mass are monitored in real time based on a predictive model, achieving precise tracking throughout the entire process (**Video S2)**.